

ESTIMATION METHODS

Nelson R. Ramírez-Rondán

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Probability, basic

- A random variable is a real-valued outcome; a function from the sample space S to the real line $\mathbb{R}$
- Let X and Y be two random variables
- $E(X) = \sum p_i X_i$ Primer momento
- $Var(X) = \sum p_i (X_i - E(X))^2$ Segundo momento
- $Var(X) = \sum [p_i X_i^2 - 2p_i X_i E(X) + E(X)^2] = E(X^2) - E(X)^2$
- $Cov(X, Y) = \sum p_i (X_i - E(X))(Y_i - E(Y))$
- $Cov(X, Y) = \sum (p_i X_i Y_i - p_i X_i E(Y) - p_i E(X) Y_i + p_i E(X) E(Y))$
- $Cov(X, Y) = E(XY) - E(X)E(Y)$
- $Cov(X, X) = E(XX) - E(X)E(X) = Var(X)$

Probability, basic

- Let X and Y be two random variables, a and b two constants.
- $E(aX + bY) = aE(X) + bE(Y)$
- $Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2abCov(X, Y)$
- Unconditional expectation: $E(X_{t+1}) = \mu$
- Conditional expectation: $E_t(X_{t+1}) \equiv E(X_{t+1} | \Omega_t)$
- Suppose X_t follows an AR(1) process $X_{t+1} = \rho X_t + \epsilon_{t+1}$ $E_t(\epsilon_{t+1}) = 0$
- $E_t(X_{t+1}) = E_t(\rho X_t + \epsilon_{t+1}) = \rho X_t + E_t(\epsilon_{t+1}) = \rho X_t$
- $E_t(X_{t+2}) = E_t(\rho X_{t+1} + \epsilon_{t+2}) = \rho E_t(X_{t+1}) + E_t(\epsilon_{t+2}) = \rho^2 X_t$
- $E_t(\epsilon_{t+2}) = E_t E_{t+1}(\epsilon_{t+2}) = E_t 0 = 0$ (law of iterated expectations)

Matrices and matrix calculus

- $[A]_{i,j} = A_{i,j}$
- $[A']_{i,j} = A_{j,i}$ (transpose)
- $(AB)' = B'A'$
- $A' = A$ if A is symmetric
- $F(x) = x^T A x$ ($1 \times n$, $n \times n$, $n \times 1$)(1×1)
- $\nabla F(x) = 2Ax$ ($n \times n$, $n \times 1$)($n \times 1$) (A is symmetric)

- $F(x) = g(x)^T A g(x)$
- $\nabla F(x) = 2A g(x)$



Least Squares Regression

The random variables $\{(Y_1, X_1), \dots, (Y_i, X_i), \dots, (Y_n, X_n)\}$ are independent and identically distributed.

The variables (Y, X) satisfy the linear regression equation

$$Y = X'\beta + e$$

$$E[e|X] = 0$$

We will consider both the general case of **heteroskedastic** regression where the conditional variance $E[e^2|X] = \sigma^2(X)$ is unrestricted.

And the specialized **case of homoskedastic** regression where the conditional variance is constant $E[e^2|X] = \sigma^2$.

Ordinary least squares (OLS) Estimator

- Linear Model $Y = X'\beta + e$ (dimensions $T \times 1$, $T \times K$, $K \times 1$, $T \times 1$)
- Assumptions
 - ▶ $E(X'e) = 0$ (first order conditions for Least Squares)
 - ▶ $E(e) = 0$
 - ▶ $E(ee') = \Omega = \text{diag}\{\sigma^2, \dots, \sigma^2\}$
 - ▶ $\text{Rank}(X) = K$ (no multi-collinearity)

Ordinary least squares (OLS) Estimator

- $Y = X'\beta + e$ $e = y - X'b$
- $\text{Min } e'e$ (minimize sum of squared errors)
- First order conditions

$$\frac{\partial e'e}{\partial \beta} = \frac{\partial (Y'Y - Y'X\beta - \beta X'Y + \beta X'X\beta')}{\partial \beta} = -2X'Y + 2X'X\beta = 0$$

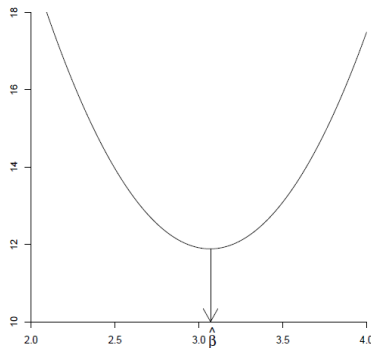
- Solving for  β , we get  $(X'X)^{-1}X'Y$
- $E(X'e) = 0$ for consistency

Ordinary least squares (OLS) Estimator

Figure 1: Regression with one regressor



(a) Deviation from Fitted Line

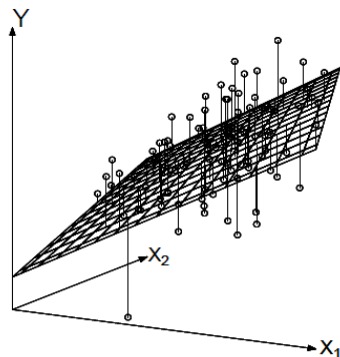


(b) Sum of Squared Error Function

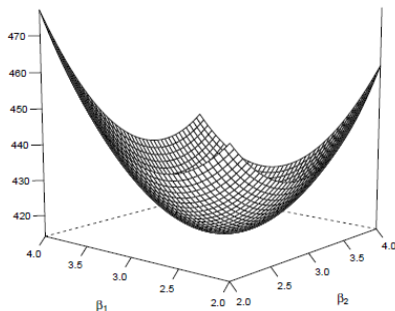
Source: Hansen (2021).

Ordinary least squares (OLS) Estimator

Figure 2: Regression with two regressors



(a) Regression Plane



(b) Sum of Squared Error Function

Source: Hansen (2021).

Ordinary least squares (OLS) Estimator, asymptotics

- Consistency. $\hat{\beta}$ converges in probability to β as the number of observations grows to infinity
- Asymptotic Normality. Asymptotic normal approximation to the distribution of the least squares estimator $\hat{\beta}$
- Unbiased
 - ▶ $\hat{\beta} = (X'X)^{-1}X'(X\beta + e)$
 - ▶ $\hat{\beta} = (X'X)^{-1}X'X\beta + (X'X)^{-1}X'e$
 - ▶ $\hat{\beta} = \beta + (X'X)^{-1}X'e$
 - ▶ $E(\hat{\beta}) = \beta + (X'X)^{-1}E(X'e)$ (conditional on X)
 - ▶ $E(\hat{\beta}) = \beta$

OLS - Best Linear Unbiased Estimator (BLUE) Unbiased

- $Y = X\beta + e$
- $\hat{\beta} = (X'X)^{-1}X'Y$
- $\hat{\beta} = (X'X)^{-1}X'(X\beta + e)$
- $\hat{\beta} = (X'X)^{-1}X'X\beta + (X'X)^{-1}X'e$
- $\hat{\beta} = \beta + (X'X)^{-1}X'e$
- $E(\hat{\beta}) = \beta + (X'X)^{-1}E(X'e)$
- $E(\hat{\beta}) = \beta$

OLS (BLUE) Best I

- $Y = X\beta + e$
- $\hat{\beta} = (X'X)^{-1}X'Y$
- $E\hat{\beta}\hat{\beta}' = E(X'X)^{-1}X'YY'X(X'X)^{-1}$
- $E\hat{\beta}\hat{\beta}' = E(X'X)^{-1}X'(X\beta + e)(X\beta + e)'X(X'X)^{-1}$
- $E\hat{\beta}\hat{\beta}' = (X'X)^{-1}X'(X\beta e' + ee' + X\beta\beta'X' + e\beta'X')X(X'X)^{-1}$
- $E(X'X)^{-1}X'(ee')X(X'X)^{-1} = \sigma^2(X'X)^{-1}$
- $E(X'X)^{-1}X'X(\beta\beta')X'X(X'X)^{-1} = \beta\beta'$

OLS (BBLUE) Best II

- $\hat{\beta} = (X'X)^{-1}X'Y$
- $E(\hat{\beta} - \beta)(\hat{\beta} - \beta)' = E(\hat{\beta} - \beta)(\hat{\beta}' - \beta')$
- $E(\hat{\beta} - \beta)(\hat{\beta} - \beta)' = E(\hat{\beta}\hat{\beta}' - \beta\hat{\beta}' + \hat{\beta}\beta' - \beta\beta')$
- $E(\hat{\beta} - \beta)(\hat{\beta} - \beta)' = E(\hat{\beta}\hat{\beta}') - \beta\beta'$
- $E(\hat{\beta}\hat{\beta}') = \sigma^2(X'X)^{-1} + \beta\beta'$
- $E(\hat{\beta} - \beta)(\hat{\beta} - \beta)' = \sigma^2(X'X)^{-1}$

OLS (BBLUE) Best III

- $\hat{\beta} = (X'X)^{-1}X'Y$
- $\hat{\Gamma} = KY$ (Linear)
where $K = (X'X)^{-1}X' + C$
- $E\hat{\Gamma} = ((X'X)^{-1}X' + C)(X\beta + e) = \beta + CX\beta$
- $E\hat{\Gamma}\hat{\Gamma}' = KEYY'K' = \sigma^2KK'$
- $E\hat{\Gamma}\hat{\Gamma}' = \sigma^2I_K((X'X)^{-1}X' + C)(X(X'X)^{-1} + C')$
- $E\hat{\Gamma}\hat{\Gamma}' = \sigma^2I_K((X'X)^{-1} + CX(X'X)^{-1} + (X'X)^{-1}X'C' + C'C)$
- $E\hat{\Gamma}\hat{\Gamma}' = \sigma^2I_K((X'X)^{-1} + C'C) > \sigma^2(X'X)^{-1}$

Maximum Likelihood Estimator

- Maximum Likelihood
- The problem is: given a probability density function and data, obtain an estimate of its parameters
- Key assumption: The distribution
- Time series models. $AR(1)$
- Maximum likelihood can be useful for estimations subject to restrictions

Maximum Likelihood - Preliminaries

- $Pr(A, B) = Pr(A)Pr(B)$
- $f(x_t|\theta)$
- $f(x_1, \dots, x_T|\theta) = f(x_1|\theta) \dots f(x_T|\theta)$
- $L(X|\theta) = f(x_1|\theta) \dots f(x_T|\theta)$
- $\max \log F$ is equivalent to $\max F$. This is, if θ^* is the solution of $\max \log F$ if and only if θ^* is the solution of $\max F$
- This, of course, does not mean that $\log F(\theta^*) = F(\theta^*)$

Maximum Likelihood I

- $f(x_1|\theta) \dots f(x_T|\theta) \equiv \prod_{t=1}^T f(x_t|\theta)$
- $\max \log F$ is equivalent to $\max F$. This is, if θ^* is the solution of $\max \log F$ if and only if θ^* is the solution of $\max F$
- $\log L(X|\theta) = l(X|\theta) = \log f(x_1|\theta) + \dots + \log f(x_T|\theta)$
- if $f(x_t|\theta) = K(\frac{1}{\sigma^2} \exp(-\frac{1}{2}(\frac{x_t - \mu}{\sigma})^2)$
- $\log f(x_t|\theta) = k - \log \sigma^2 - \frac{1}{2}(\frac{x_t - \mu}{\sigma})^2$
- $\log L(X|\theta) = Tk - T \log \sigma^2 - \sum \frac{1}{2}(\frac{x_t - \mu}{\sigma})^2$

Maximum Likelihood II

- $\log L(X|\theta) = Tk - T \log \sigma^2 - \sum \frac{1}{2} \left(\frac{x_t - \mu}{\sigma} \right)^2$

First order conditions

- $[\mu] \quad - \sum \left(\frac{x_t - \mu}{\sigma} \right) \frac{1}{\sigma} = 0$

- $[\mu] \quad \mu = \frac{1}{T} \sum x_t$

- $[\sigma] \quad \frac{T}{\sigma^2} 2\sigma - \frac{2}{2} \sum \left(\frac{x_t - \mu}{\sigma} \right) \left(\frac{x_t - \mu}{\sigma^2} \right) = 0$

- $[\sigma^2] \quad \sigma^2 = \frac{1}{T} \sum (x_t - \mu)^2$

Maximum Likelihood III

Let the model specified before

$$y_t = \beta x_t + \epsilon_t$$

$$E[\epsilon_t | x_t] = 0$$

$$\epsilon_t \sim N(0, \sigma^2)$$

Then,

$$\hat{\beta} = \underset{\beta}{\operatorname{argmax}} L = \underset{\beta}{\operatorname{argmax}} \prod_{t=1}^T \left[\frac{1}{2\pi\sigma^2} \exp \left(-\frac{(y_t - \beta x_t)^2}{2\sigma^2} \right) \right]$$

$$\hat{\beta} = \underset{\beta}{\operatorname{argmax}} \ln L = \underset{\beta}{\operatorname{argmax}} \left[-\frac{T}{2} \ln(2\pi) - \frac{T}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^T (y_t - \beta x_t)^2 \right]$$

Maximum Likelihood IV

- $x_{t+1} = Ax_t + \epsilon_{t+1}$ where ϵ_{t+1} distributes iid $N(0, \Omega)$
- $f(x_{t+1}|\theta) = K(\frac{1}{|\Sigma|})^{1/2} \exp(-\frac{1}{2}(x_{t+1} - Ax_t)' \Sigma^{-1}(x_{t+1} - Ax_t))$
- $\max \log F$
- s.t. $A_{1,1} = 0$

Maximum Likelihood IV

- $x_{t+1} = Ax_t + \epsilon_{t+1}$ where ϵ_{t+1} distributes iid $N(0, \Omega)$
- $f(x_{t+1}|\theta) = K(\frac{1}{|\Sigma|})^{1/2} \exp(-\frac{1}{2}(x_{t+1} - Ax_t)' \Sigma^{-1}(x_{t+1} - Ax_t))$
- $\max \log F$
- Global optimal
- Variance-Covariance matrix: $\text{Var}(\theta) = (-E[\nabla^2 \log L])^{-1}$

Maximum Likelihood - Tests

- Model Selection Criteria
- AIC Akaike information criteria: $2K - 2\log(L)$
- HQIC Hannan-Quinn information criterion: $2K \log \log(T) - 2\log(L)$
- SBIC Schwarz Bayesian information criterion: $K \log(T) - 2\log(L)$

where L is the likelihood function optimal value, K is the number of parameters, and T is the number of observations.

Endogeneity

Let the linear Model

$$Y = X'\beta + e$$

$$E(X'e) \neq 0$$

In this case, we say that X is endogenous for β , where $\hat{\beta} \neq \beta$.

To consistently estimate β we require additional information. One type of information which is commonly used in economic applications are what we call **instruments**.

A random variable Z is an instrumental variable if $E(Ze) = 0$ and it is correlated with X .

GMM Motivation

- It is a general estimation method
- Several estimators can be seen as special cases of GMM
- It makes minimal assumptions
- For instance, by using MLE, one assumes a lot more, notably, the probability density function.
- It is plausible when others might fail
- One does not need to specify all the model

- $E[g_t(\beta)] = 0$ Theoretical equation
- For example, in the instrumental variables model $g_t(\beta) = Z_t(Y_t - X_t'\beta)$, where Z_t is the instrumental variable (no correlated with the error term)
- $\bar{g}_t(\beta) \equiv \frac{1}{T} \sum_{t=1}^T g_t(\beta)$ This is the empirical counterpart
- K number of parameters, N number of moment conditions, T number of observations.
- The relationship between K and N is key to a GMM estimation
 - ▶ If $N < K$ the system is under-identified (not feasible)
 - ▶ if $N = K$ the system is identified (feasible)
 - ▶ If $N > K$ the system is over-identified (feasible)

GMM - Estimation

In particular, we focus on the over-identified IV model with moment equations

$$g_t(\beta) = Z_t(Y_t - X_t'\beta)$$

the sample moment equations are

$$\bar{g}_t(\beta) = \frac{1}{T} \sum_{t=1}^T g_t(\beta) = \frac{1}{T} \sum_{t=1}^T Z_t(Y_t - X_t'\beta) = \frac{1}{T}(Z'Y - Z'X\beta)$$

The GMM criterion can be written as

$$J(\beta) = T(Z'Y - Z'X\beta)'W(Z'Y - Z'X\beta)$$

The GMM estimator minimizes $J(\beta)$.

GMM - Estimation

The first order conditions are

$$0 = \frac{\partial J}{\partial \beta} J(\hat{\beta})$$

$$0 = \frac{\partial J}{\partial \beta} \bar{g}_t(\hat{\beta})' W \bar{g}_t(\hat{\beta})$$

$$0 = -2 \left(\frac{1}{T} X' Z \right) W \left(\frac{1}{T} Z' (Y - X \hat{\beta}) \right)$$

The solution is given as follows

$$\hat{\beta}_{GMM} = (X' Z W Z' X)^{-1} (X' Z W Z' Y)$$

While the estimator depends on the weight matrix W the dependence is only up to scale. When W is fixed we call $\hat{\beta}$ a **one-step GMM** estimator.

GMM - Estimation

When the model is just-identified ($N = K$) then $X'Z$ is $K \times K$ so the expression for the estimator simplifies to the IV estimator

$$\hat{\beta}_{GMM} = (Z'X)^{-1}W^{-1}(X'Z)^{-1}(X'ZWZ'Y) = (Z'X)^{-1}(Z'Y) = \hat{\beta}_{IV}$$

When $W = (Z'Z)^{-1}$, then the GMM estimator is the two-stage-least squares estimator:

$$\hat{\beta}_{GMM} = (X'Z(Z'Z)^{-1}Z'X)^{-1}(X'Z(Z'Z)^{-1}Z'Y) = \hat{\beta}_{2SLS}$$

GMM - Asymptotic Distribution

Asymptotic Distribution of GMM Estimator

as $T \rightarrow \infty$ then $\sqrt{T}(\hat{\beta}_{GMM} - \beta) \rightarrow_d N(0, V_\beta)$ where

$$V_\beta = (Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1}$$

given $Q = E[ZX']$ and $\Omega = E[ZZ'\epsilon^2]$

When the GMM estimator $\hat{\beta}$ is constructed with $W = \Omega^{-1}$, we call it the **Efficient GMM** estimator

$$\hat{\beta}_{GMM} = (X'Z\Omega^{-1}Z'X)^{-1}(X'Z\Omega^{-1}Z'Y)$$

And the asymptotic variance is $V_\beta = (Q'\Omega^{-1}Q)^{-1}$.

GMM - Extensions

- Two-step GMM estimator
- Iterated GMM
- Restricted GMM
- Nonlinear Restricted GMM
- Continuously-Updated GMM
- Over-identification Test
- Nonlinear GMM

GMM and OLS

Let the linear equation

$$Y = X'\beta + \epsilon$$

$$E[\epsilon|X] = 0 \Rightarrow E[X'\epsilon] = 0$$

Which implies,

$$E(X'(Y - X\beta)) = 0$$

$$E(X'Y - X'X\beta) = 0$$

$$E(X'Y) = E(X'X)\beta$$

$$\beta = E(X'Y)^{-1}E(X'X)$$

$$\hat{\beta} = (X'X)^{-1}X'Y$$